

Supplementary Material S2: Complete Search Strategies

PROSPERO Registration

CRD420261354170

General Search Parameters

- **Date range:** 1 January 2000 to 27 March 2026
- **Language restriction:** English
- **Search date:** 28-29 March 2026

The search strategy combined three conceptual domains: 1. **Ultrasound guidance** (MeSH/Emtree terms + free text) 2. **Peripheral intravenous catheterisation** (MeSH/Emtree terms + free text) 3. **Post-insertion outcomes** (MeSH/Emtree terms + free text)

1. PubMed (n = 530)

Search date: 28 March 2026

("Ultrasonography"[MeSH] OR "ultrasound"[tiab] OR "ultrasonograph*"[tiab]
OR "sonograph*"[tiab] OR "POCUS"[tiab] OR "point-of-care ultrasound"[tiab]
OR "echograph*"[tiab] OR "US-guided"[tiab] OR "USG"[tiab]
OR "ultrasound-guided"[tiab] OR "echo-guided"[tiab])

AND

("Catheterization, Peripheral"[MeSH] OR "peripheral intravenous"[tiab]
OR "peripheral venous catheter*"[tiab] OR "peripheral IV"[tiab]
OR "PIVC"[tiab] OR "PIV"[tiab] OR "peripheral vascular access"[tiab]
OR "peripheral venous access"[tiab] OR "peripheral cannulat*"[tiab]
OR "intravenous cannulat*"[tiab] OR "venipuncture"[tiab]
OR "venepuncture"[tiab] OR "short peripheral catheter"[tiab]
OR "USGIV"[tiab] OR "USGPIV"[tiab] OR "USG-PIVC"[tiab])

AND

("Equipment Failure"[MeSH] OR "Phlebitis"[MeSH]
OR "Catheter-Related Infections"[MeSH] OR "dwell time"[tiab]
OR "catheter survival"[tiab] OR "catheter failure"[tiab]
OR "catheter longevity"[tiab] OR "catheter durability"[tiab]
OR "catheter complication*"[tiab] OR "post-insertion"[tiab]
OR "phlebitis"[tiab] OR "infiltration"[tiab] OR "extravasation"[tiab]
OR "occlusion"[tiab] OR "dislodgement"[tiab] OR "dislodgment"[tiab]
OR "catheter patency"[tiab] OR "catheter-related infection"[tiab]
OR "thrombophlebitis"[tiab] OR "premature removal"[tiab]
OR "unplanned removal"[tiab] OR "catheter-related bloodstream"[tiab]
OR "device failure"[tiab] OR "catheter malfunction"[tiab])

AND

("2000/01/01"[Date - Publication] : "2026/03/27"[Date - Publication])

2. Embase via Ovid (n = 984)

Search date: 29 March 2026

Line	Search
1	exp echography/ OR ultrasound.tw. OR ultrasonograph*.tw. OR sonograph*.tw. OR POCUS.tw. OR "point-of-care ultrasound".tw. OR "US-guided".tw. OR USG.tw. OR "ultrasound-guided".tw.
2	exp peripheral catheterization/ OR "peripheral intravenous".tw. OR "peripheral venous catheter*".tw. OR PIVC.tw. OR PIV.tw. OR "peripheral vascular access".tw. OR "peripheral venous access".tw. OR "peripheral cannulat*".tw. OR "intravenous cannulat*".tw. OR venipuncture.tw. OR venepuncture.tw. OR "short peripheral catheter".tw. OR USGIV.tw. OR USGPIV.tw.
3	exp device failure/ OR exp phlebitis/ OR exp catheter infection/ OR "dwell time".tw. OR "catheter survival".tw. OR "catheter failure".tw. OR "catheter complication*".tw. OR phlebitis.tw. OR infiltration.tw. OR extravasation.tw. OR occlusion.tw. OR dislodg*.tw. OR "catheter patency".tw. OR thrombophlebitis.tw. OR (premature adj2 removal).tw. OR (unplanned adj2 removal).tw. OR "device failure".tw. OR "catheter malfunction".tw.
4	1 AND 2 AND 3
5	limit 4 to yr="2000-2026"
6	5 NOT (exp animal/ NOT exp human/)

3. Cochrane Central Register of Controlled Trials (CENTRAL) (n = 68)

Search date: 29 March 2026 Platform: Cochrane Library Advanced Search (Trials tab)

Line	Search
#1	MeSH descriptor: [Ultrasonography] explode all trees OR (ultrasound OR ultrasonograph* OR sonograph* OR POCUS OR "point-of-care ultrasound" OR "US-guided" OR "USG" OR "ultrasound-guided"):ti,ab,kw
#2	MeSH descriptor: [Catheterization, Peripheral] explode all trees OR ("peripheral intravenous" OR "peripheral venous catheter*" OR "peripheral IV" OR PIVC OR PIV OR "peripheral vascular access" OR "peripheral venous access" OR "peripheral cannulat*" OR "intravenous cannulat*" OR venipuncture OR venepuncture OR "short peripheral catheter" OR USGIV OR USGPIV):ti,ab,kw
#3	MeSH descriptor: [Equipment Failure] explode all trees OR MeSH descriptor: [Phlebitis] explode all trees OR MeSH descriptor: [Catheter-Related Infections] explode all trees OR ("dwell time" OR "catheter survival" OR "catheter failure" OR "catheter longevity" OR "catheter durability" OR "catheter complication*" OR "post-insertion" OR phlebitis OR infiltration OR extravasation OR occlusion OR dislodgement OR dislodgment OR "catheter patency" OR thrombophlebitis OR "premature removal" OR "unplanned removal" OR "device failure" OR "catheter malfunction"):ti,ab,kw
#4	#1 AND #2 AND #3
#5	#4 with Publication Year from 2000 to 2026

4. CINAHL Complete via EBSCOhost (n = 50)

Search date: 29 March 2026

Line	Search
S1	(MH "Ultrasonography+") OR TI ultrasound OR AB ultrasound OR TI ultrasonograph* OR AB POCUS OR TI "point-of-care ultrasound" OR TI "ultrasound-guided" OR AB "US-guided"
S2	(MH "Catheter Insertion, Peripheral") OR (MH "Peripheral Vascular Devices") OR TI "peripheral intravenous" OR AB "peripheral venous catheter*" OR TI PIVC OR AB "peripheral cannulat*" OR TI "intravenous cannulat*" OR AB venipuncture
S3	(MH "Phlebitis") OR (MH "Equipment Failure") OR (MH "Catheter-Related Infections") OR TI "dwell time" OR AB "catheter failure" OR TI phlebitis OR AB infiltration OR TI extravasation OR AB occlusion OR TI dislodg* OR AB "catheter complication*" OR TI "catheter survival" OR AB "premature removal" OR TI "unplanned removal" OR AB thrombophlebitis
S4	S1 AND S2 AND S3
S5	S4 (Limiters: Published Date 2000-2026)

Search Results Summary

Database	Records retrieved
PubMed	530
Embase (Ovid)	984
Cochrane CENTRAL	68
CINAHL (EBSCOhost)	50
Total	1,632
Duplicates removed	273
Unique records screened	1,359

Deduplication Method: Three-layer deduplication was performed using Python: 1. **DOI matching** (exact) 2. **PMID matching** (exact) 3. **Title similarity** (Jaccard coefficient ≥ 0.85)